

SMASH - MISSION 2 PRODUCTION GUIDE

Beach / seaside. What the mission is for, how to build it, what will break.

Supersedes section 3 of SMASH_Mission2_3_Level_Blueprint.txt, which was written before the coating system, the panel/weld rule and the Level 8 build.

Reads with: SMASH_MapJSON_Spec_and_Review, SMASH_JSON_Cloning_Handbook, SMASH_WallCoating_Spec

Project: Demolition Contracts (FierceTigerGameJam) Revision 1 - 2026-08-30

WHO THIS IS FOR

Whoever produces the nine Mission 2 maps - concept images and JSON. Section 1 is the argument for what the mission is; skipping it produces nine good-looking buildings that play exactly like Mission 1.

1. WHAT MISSION 2 IS FOR - THE ONE IDEA

MISSION 1 TAUGHT ATTRITION.

Nine closed boxes. Hollow shells with windows, but closed: the mass sits on the ground and the way through it is to grind it down. The player learns hit points, materials, and that bricks are cheaper alone than in a wall.

MISSION 2 MUST TEACH LEVERAGE.

Beach architecture is naturally OPEN, RAISED AND LIGHT: huts on stilts, piers on pile bents, bars with no front wall, slides cantilevered off a tower, a lighthouse that is a tall thin shaft. Not one of those is solved by attrition. Every one of them is solved by finding the SUPPORT and taking it out. That is the mission's lesson, and the theme hands it over for free.

THE NUMBERS ALREADY PUSH THE PLAYER THERE - use it.

Per SMASH_WallCoating_Spec, Mission 2 has BRICK COATED on every level. On Rock Lvl a coated brick wall takes 1 damage per shot where a loose brick takes 3. Grinding a Mission 2 wall is three times as slow as grinding a Mission 1 wall. Meanwhile a leg is 4-6 cells: cheap to cut, and the mass it carries costs nothing to destroy because gravity does it. Design and tuning agree. Attrition is expensive; leverage is free. The player will find that out whether or not anyone tells them.

WHAT THIS MEANS PER LEVEL

Every Mission 2 map must have an answer to: WHICH 4-8 CELLS, IF REMOVED, BRING DOWN MORE THAN THEIR OWN WEIGHT? If a map has no such answer it is a Mission 1 map wearing a beach hat - send it back.

THE ONE THING NOT TO REPEAT

No more four-walled closed boxes. Mission 1 has nine. At most two Mission 2 levels may be closed volumes (LV7 lighthouse shaft, LV8 fort casemate) and both must earn it by being tall or by hiding an interior worth reaching.

2. THEME WITHOUT NEW ART - PAINTING THREE BLOCKS AS A BEACH

There are four block types and no more, ever: brick_1x1, brick_2x1, glass_1x1, concrete_1x1. The beach is carried by SHAPE and by how the concept image paints the same three materials.

brick	sand-brick, bleached timber, decking. The mission's body.
glass	sea-view glazing, wind screens, slide tubes, lantern rooms.
concrete	stone piers, breakwater, rock plinths, the lighthouse base.

SILHOUETTE IS THE WHOLE THEME. What makes a beach read is: things standing on legs above a flat ground plane, long horizontal decks, open fronts, and one tall slim thing on the horizon. Get those and the palette follows.

COLOUR NOTE FOR THE IMAGE GENERATOR: render at 1 brick = 1 cell of the grid you intend to emit (spec section 0.1). A concept drawn at finer resolution than the grid is a promise the JSON cannot keep - this has already cost one rejected Lifeguard Tower.

3. THE STILT TRAP - READ THIS BEFORE AUTHORIZING ANY RAISED STRUCTURE

The engine releases a body when the thing DIRECTLY BELOW ITS FOOTPRINT is destroyed. KnockdownBlock.IsSupportedAbove tests overlap in x AND z, then releases whatever sits over that footprint.

A body held only from the SIDE is never released by anything.

Nothing below it will ever fall, so nothing will ever tell it to fall. It hangs in the air forever and blocks the 100% clear. Mission 2 is the mission where this bites, because half its buildings are raised.

- R1 EVERY RAISED MASS MUST HAVE AT LEAST ONE CELL SITTING DIRECTLY ON A LEG.
A cabin on four legs: the cabin floor must include the four cells directly above the leg tops. A pier deck: every bent must have deck cells directly over its posts.
- R2 A CANTILEVER MUST BE ANCHORED THROUGH A CELL ABOVE ITS PROP, not merely touching the tower beside it. The waterslide ribbon (LV6) is the risky one: give it real props at intervals, not a lateral attachment.
- R3 RUN THE VALIDATOR AND READ THE FOOTING LINES. Zero unfooted walls and zero floating loose blocks, every time. All nine Mission 1 maps pass; so must these.
- R4 A LEG IS ITS OWN WALL. One wall per leg, 4-6 cells. Then one shot falls exactly one leg, which is the whole point. A leg merged into the floor slab cannot be cut.

4. WHAT LEVEL 8 TAUGHT - CARRY THESE OVER

The Village Gatehouse went through four rebuilds. The lessons are cheap to inherit and expensive to rediscover.

- 4.1 VERTICAL RUNS MAKE COLUMNS, AND COLUMNS ARE THE PRETTIEST THING WE HAVE
A 1-cell-wide vertical stack of concrete authored as its own wall welds into one continuous stanchion. It reads as a structural column, not a pile of cubes. Use it for every stilt, post, pile and pier leg in this mission - it is the theme's signature AND the R4 rule above, for free.
- 4.2 ONE-CELL OPENINGS, NOT TWO
A tower face four cells wide with concrete columns at both ends has two cells of brick between them. Punch a two-wide hole and the entire face above it is stranded. One-wide slits keep a continuous strip running to the ground. In beach terms: tall narrow shutters, not picture windows - except where the whole storey is glass, which is a different move.
- 4.3 NEVER MERGE A TWO-CELL WALL INTO ONE brick_2x1
The merge leaves the wall holding a single entry, and the engine treats a one-block wall as an ordinary loose block - quietly undoing the grouping. Guard the merge on wall size > 2.
- 4.4 MIND WHAT THE FOOTING REPAIR CANNOT FIX
The repair pass merges an unfooted wall into a touching neighbour OF THE SAME MATERIAL. A brick row sitting between two concrete columns with a hole underneath has no same-material neighbour and stays broken. Fix the geometry, do not relax the pass.

4.5 A HEAVY LID ON A GLASS BELT IS THE BEST SET PIECE IN THE GAME

Level 8's concrete crown sits on a 1 HP glass band. One shot into the glass drops the entire ring beam onto the brick below. Beach equivalents: the bar roof on glass wind-screens (LV4), the restaurant roof deck on the glass dining storey (LV5). Use it, but once or twice, not nine times.

5. CONTROLLING HOW A WALL LOOKS - PANEL vs WELDED

A wall is drawn one of two ways, and the map decides which without meaning to.

PANEL (smooth, poured, one stretched slab) is used when ALL of:

- one material, all cells 1x1, the wall lies in ONE layer (size.z == 1),
- and it fills a solid rectangle with no holes.

WELDED (individual blocks visible) in every other case.

Neither is better. They just have to be chosen on purpose:

- smooth suits columns, stone plinths, glass curtains, slide tubes

- blocky suits masonry, decks, parapets, anything meant to read as laid

TO FORCE BLOCKY: give the wall a hole, or group it so it spans more than one layer. Level 8's crown was two flat 10x1x1 runs (smooth); grouping the whole ring together made it span every layer and it welds.

TO FORCE SMOOTH: author a solid single-layer rectangle - which is exactly what a 1xN column is.

There is also a global useWallPanels switch on the authoring component, but it is all-or-nothing; prefer shaping the group.

6. THE BUILD SHEET - NINE LEVELS

Grid, entries and the collapse story stay as the Blueprint set them. Added here:

the coating per level, the support target, and what must not go wrong.

Coating follows SMASH_WallCoating_Spec section 3 - Mission 2 has BRICK COATED on every level, CONCRETE COATED from LV7, glass never counted.

LV1 BEACH HUT ROW brick 14x6x5 ~110 entries

Three tiny huts, different roofs, each on short stilts.

SUPPORT TARGET: a hut's two stilts - it slides into its neighbour.

COATING: brick armoured. No concrete, no glass.

WATCH: each hut is its own set of walls; do not let the optimizer weld two huts into one body across the gap between them.

LV2 LIFEGUARD TOWER brick 8x11x6 ~130 entries

Cabin with peaked roof on four tall legs, ramp leaning against it.

SUPPORT TARGET: two legs and the cabin belly-flops.

COATING: brick armoured.

WATCH: the cabin floor MUST include the four cells over the leg tops (R1). The ramp is one cell deep and is a separate body. Concept image already reviewed - v2 approved, with the legs honest at 6 rows.

LV3 THE PIER brick 10x7x11 ~160 entries

Boardwalk running away from the player on four pile bents, kiosk at the end. The mission's depth level.

SUPPORT TARGET: the front bent - the deck pancakes bent by bent.

COATING: brick armoured.

WATCH: deck cells must sit directly over each bent's posts (R1), or the far half of the pier hangs after the near half falls. This level is the single most likely place in the mission to produce a floating deck.

LV4 BEACH BAR brick+glass 12x7x7 ~170 entries

Open front, heavy flat roof on two posts, glass wind-screens at the sides, loose bottle blocks on the counter.

SUPPORT TARGET: one front post - the roof slaps down on the counter.
COATING: brick armoured, glass loose-grouped per screen.
WATCH: first use of 4.5 - the roof drops through the glass. The two posts are separate 1xN walls (4.1). Bottles are loose 1x1s, 6-10 of them.

LV5 SEAVIEW RESTAURANT brick+glass 12x9x7 ~200 entries
Solid brick ground floor, one FULL GLASS STOREY, brick roof deck with loose parasol stubs.
SUPPORT TARGET: the glass storey itself - the classic waist.
COATING: brick armoured; the glass storey is one or two big walls so the whole dining room goes at once.
WATCH: do not put a brick pier through the glass storey "to be safe" - it kills the move. The roof deck is allowed to be held only by glass.

LV6 WATERSLIDE TOWER brick+glass 14x11x7 ~220 entries
Stair tower, stepped diagonal glass tube descending to a pool rim, separate ticket kiosk.
SUPPORT TARGET: the slide's props.
COATING: brick armoured.
WATCH: the highest-risk level in the mission for R2. The tube is a diagonal run; every 2-3 cells of it needs a prop underneath, and each prop is its own 1xN wall. A tube attached only to the tower's flank will hang forever. Consider proving this one in the Smash Builder gameplay tab before delivering.

LV7 LIGHTHOUSE all three 9x14x6 ~230 entries
Concrete rock base, tapering brick shaft stepping in twice, glass lantern room, keeper's cottage at the foot.
SUPPORT TARGET: one side of the shaft - the whole tower rides down onto the cottage.
COATING: brick AND CONCRETE armoured from here on.
WATCH: FIRST CANNON-GATED LEVEL. Rock does 0 damage to concrete at every level, so the concrete base cannot be cleared with the starting ammunition at all. Either the base is small enough to be knocked down by the collapsing shaft rather than shot, or the level is explicitly a Cannon wall. Decide deliberately and write it in the manifest.

LV8 SEA FORT all three 12x10x8 ~270 entries
Squat concrete casemate on a stone plinth, brick barracks in the open court, glass observation slit band, loose battlement caps.
SUPPORT TARGET: none - this is the mission's one anatomy level. The player shoots through the slit band and guts the brick interior.
COATING: brick and concrete armoured.
WATCH: the concrete shell must not enclose the brick so completely that a Rock player can do nothing. The slit band is the way in and must be wide enough to shoot through at a shallow angle.

LV9 BOARDWALK RESORT (BOSS) all three 16x12x10 ~340 entries
Lifeguard tower + beach bar + short pier stub sharing one beach.
SUPPORT TARGET: three of them, and the pier deck can be dropped onto the bar's roof.
COATING: brick and concrete armoured.
WATCH: entries and loose bodies both near the ceiling. Keep loose under 150 and total under 400. Three structures means three separate wall families - nothing may weld across the gaps between them.

BUDGET REMINDERS

entries by band: tutorial 50-120, mid 120-220, hard 220-350, boss 280-400
loose bodies <= 150 on the opening frame, always
wall size 4-24 cells, hard bounds 2 and 30
grid <= 16 wide, 14 tall, 12 layers

7. ORDER OF WORK

1. Concept image at true cell resolution, one level at a time.
2. Review against section 3 and 6 - is there a support target? does every raised mass sit over a leg? is it within grid and budget?
3. Build the JSON. Legs and posts as 1xN single-material walls; masses by face and storey; brick_2x1 merge only where the wall is larger than 2.
4. Coating: set nothing on brick (armoured is the default); set "bare": true on glass everywhere, and on concrete for LV1-6 - there is no concrete before LV7, so in practice only glass carries the flag until the lighthouse.
5. Validate. Zero errors, and read the footing lines specifically.
6. Voxel-render the JSON and put it beside the concept. The JSON ships.
7. Play it in the Smash Builder "Review gameplay" tab. Confirm the support target actually works and that nothing is left hanging at the end.
8. Deliver .json + render + manifest line: id, grid, entries, walls, loose, materials, coating, support target, Cannon-gated yes/no.

8. CHECKLIST PER LEVEL

- [] there is a named support target of 4-8 cells that brings down more than its own weight
- [] every raised mass has a cell directly over a leg (R1); no body is held only from the side (R2)
- [] each leg / post / pile is its own 1xN wall (R4, 4.1)
- [] no two-wide holes stranding a face (4.2); no 2-cell wall merged to one entry (4.3)
- [] closed four-walled box? only LV7 and LV8 are allowed one
- [] panel/weld chosen on purpose per wall (section 5)
- [] brick armoured; glass bare; concrete armoured from LV7, bare before
- [] validator 0 errors, 0 unfooted, 0 floating
- [] entries in band, loose <= 150, walls 4-24 cells
- [] Cannon-gating decided and written down (LV7-9)
- [] played once in Review gameplay, ends at 100%

END